

"For example, while talking on a cellphone, the electronics for FM radio is completely turned off, the camera section is turned off, everything that is not in use is turned off in sections. This is done using power management IC," adds Anant.

Rushabh Solani, sales engineer, Infineon, shares "Power saving for portable devices can be achieved with low-power-consuming electronics, low standby power, good conversion efficiency, power factor control and energy-efficient displays. For non-portable devices like washing machines, power saving can be brought in through efficient motor control."

Power management ICs

Power management integrated circuits can incorporate multiple functions for power management. "For power management, there is a host controller inside every product. It is basically

Power Management IC Vendors

- Analog Devices
- Fairchild Semiconductor
- Fujitsu
- Infineon Technologies
- Intersil
- Linear Technology
- Maxim
- Microchip
- National Semiconductor
- NXP Semiconductors
- ON Semiconductor
- ST Microelectronics
- Texas Instruments
- Vishay

a low-power MCU. Also known as housekeeping equipment, it consumes much less power. Power management is done by the power management host processor, firmware and associated power management digital hardware," explains Anant.

The software helps you do event management inside the IC itself. For instance, while configuring which part of hardware you want to keep alive and what you want to turn off, you have power management software. In a laptop or a cellphone, you have the option to turn off the screen when idle for a given number of minutes or seconds. So the power management can be user-driven or system-driven.

PWM controller

"Initially, the world of power management was confined to linear power supplies, which are 30-40 per cent efficient," says Anant. These were bulky, hot and unmanageable. "Then came switch-mode power supplies with introduction to the pulse-frequency modulation (PFM) controller and pulse-width modulation (PWM) controller with hysteric control topology," he adds. Switch-mode DC-DC con-

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